

# Design of an ML-Based Predictive Maintenance System for Industrial Motors Using IoT Sensor Data

This document specifies a proposed production-oriented ML system. It is not evidence of a trained or deployed system, and it reports no experimental or operational results.

Throughout the document, **FACT** denotes a claim supported by a cited source or fixed project input, **DESIGN ASSUMPTION** denotes a provisional value introduced because plant-specific evidence is unavailable, and **DESIGN DECISION** denotes an engineering choice made for this design.

## 1. Problem Definition

### 1.1 Context and Background

Industrial motors support production processes whose interruption can create downtime and emergency maintenance work. Condition-based maintenance is not only a modelling problem: it connects data acquisition, data processing, and maintenance decision-making [1]. Motor condition can be observed through several signal families, including electrical current, vibration, speed, and thermal measurements [2]. In rotating bearings, vibration can contain fault-related impulsive content, while shaft speed affects characteristic frequencies and may require speed-aware preprocessing [3]. These findings justify a multimodal design, but they do not establish that every signal is equally useful for every motor or plant.

The organizational context is a **DESIGN ASSUMPTION**: a hypothetical industrial plant operates multiple instrumented motors and can associate sensor streams with stable asset identities. No real organization, fleet, baseline, or deployment is claimed. The proposed system covers the full lifecycle from OT data collection to a maintenance decision workflow, including ingestion, validation, training, evaluation, deployment, monitoring, feedback, and controlled retraining. It remains advisory and introduces no command path to motor actuation.

The design matters because a standalone fault classifier would not address whether data are current and trustworthy, whether alerts reach maintenance staff, whether repeated warnings cause unnecessary work, or whether a changed model can be deployed safely. The intended business value is therefore to reduce unplanned motor downtime and emergency maintenance burden while avoiding unnecessary interventions and excessive false alarms.

## 1.2 Measurable Problem Statement and Scope

For each eligible industrial motor and valid recent sensor window, the system shall use **one primary predictive model** to estimate the probability that the motor will fail within the **next 24 hours**. It shall return Normal or High Risk and the failure-risk probability. The primary inputs are recent vibration, temperature, electrical-current, and rotational-speed (RPM) measurements. RPM also provides operating context for speed-aware processing.

The target is binary classification, not remaining-useful-life regression, generic anomaly detection, or multi-class fault diagnosis. The probability supports prioritization; the class and alert episode support an operational maintenance decision. A valid result is measured against approved 24-hour failure events. System value is evaluated prospectively through downtime, maintenance, alert-delivery, latency, availability, data-pipeline, and model-quality metrics defined in Section 3. None of those metrics is claimed as achieved.

The **DESIGN ASSUMPTION** used to make the system testable is an initial population of 100 motors, with capacity validation at 500 motors. Eligible motors are scored every 15 minutes and on authenticated demand using a provisional six-hour lookback. These values are engineering planning inputs, not facts about an existing plant.

## 1.3 Current Situation

No current plant workflow was supplied. The following as-is description is therefore a **DESIGN ASSUMPTION**, not an observed organizational fact. Maintenance personnel currently combine planned maintenance, reactive work after alarms or failures, manual inspection, and deterministic sensor or equipment-protection thresholds. Work orders or failure records are assumed to capture at least motor identity and event time, although the authoritative system and the operational definition of failure remain to be approved.

This assumed workflow has three relevant limitations. First, fixed thresholds and manual inspection may not combine recent patterns across vibration, temperature, current, and RPM. Second, reactive records may be inconsistent or delayed, weakening label quality and post-event evaluation. Third, the absence of an integrated prediction lifecycle prevents systematic traceability from sensor evidence to prediction, alert disposition, maintenance action, and outcome. The proposed system supplements rather than removes deterministic equipment protection and expert judgment.

## 1.4 Justification for ML

ML is appropriate as a candidate to test because recent multimodal measurements may contain interactions and temporal patterns that a single universal threshold does not represent. This is a hypothesis to evaluate on representative plant histories, not an established property of the hypothetical plant. Predictive-maintenance research uses several ML families and emphasizes selecting a method according to the available data and operational problem rather than assuming a universally best algorithm [4]. A learned classifier can combine engineered trends, time/frequency information, RPM context, and

missingness indicators; whether its probability is sufficiently calibrated must be demonstrated through MM-06 rather than assumed.

ML is not justified merely because sensor data exist. A deterministic maintenance-approved threshold rule and logistic regression remain mandatory baselines. The candidate is valuable only if it improves prevalence-aware ranking and probability quality while meeting recall, precision, false-positive, latency, reliability, and maintainability gates. If representative plant histories and trustworthy timestamped outcomes are unavailable, or if the candidate cannot outperform these baselines, deployment is not justified.

The Paderborn dataset contains synchronous vibration and motor-current measurements with supporting operating variables [5], and CWRU provides documented motor-bearing vibration data [6]. Both can support ingestion, signal-processing, and evaluation-code prototypes. Neither dataset provides the complete production population or a native “failure within the next 24 hours” target. Their reported results must not be presented as performance of this system, and unrelated records must not be merged into artificial multimodal examples.

## 1.5 Stakeholders and Concerns

| Stakeholder                                    | Goals and decisions                                                                                                                                          | Principal concerns                                                                                                                                              |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Maintenance Engineer                           | Review prioritized motors; inspect probability, trends, and data freshness; acknowledge, assign, investigate, close, and link alerts to work orders.         | Missed failures, excessive alerts, insufficient diagnostic context, stale data shown as healthy, and inability to explain or audit a prediction.                |
| Plant Operations Manager                       | Reduce unplanned downtime and emergency work while controlling total maintenance and infrastructure cost; approve operational policy and pilot continuation. | Production disruption, unnecessary preventive work, unreliable service, unsafe automation, weak business attribution, and uncontrolled OT connectivity.         |
| ML / Platform Engineer                         | Operate ingestion, validation, training, serving, monitoring, registry, deployment, and rollback; produce reproducible evidence for promotion.               | Data/label quality, training-serving skew, latency, availability, security, drift, delayed labels, model/version traceability, and maintainable infrastructure. |
| Security / Identity Administrator (supporting) | Approve service/user identities, network boundaries, secrets, and                                                                                            | OT/IT exposure, excessive privilege, credential failure, unaudited release actions,                                                                             |

| Stakeholder                       | Goals and decisions                                                                                        | Principal concerns                                                                                                                        |
|-----------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | privileged-access controls.                                                                                | and controls that disrupt plant availability.                                                                                             |
| Data / Label Steward (supporting) | Resolve ambiguous maintenance events and approve the versioned failure/label policy and training outcomes. | Incorrect asset/time joins, inconsistent failure definitions, leakage, censored interventions, and unreviewed feedback entering training. |

Maintenance and operations must jointly approve the failure definition, alert policy, and intervention workflow. The supporting roles enforce controls but do not become additional business owners. The system remains human-in-the-loop because no actuator authority or safety case has been supplied.

## 2. Requirements Analysis

### 2.1 Requirement Status and Planning Envelope

All numerical values in this section are **DESIGN ASSUMPTIONS** or **PROPOSED DESIGN TARGETS** unless a citation states otherwise. They create measurable acceptance conditions and must be reviewed after a sensor/site survey and shadow-mode pilot.

| Planning item         | Provisional value and justification                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fleet                 | 100 motors initially; capacity test at 500 motors to test fivefold growth.                                                                                                                                                                                                                                                                                                                                                                                 |
| Scoring               | Every 15 minutes per eligible motor, plus authenticated on-demand scoring; timely relative to the 24-hour advisory horizon without implying real-time control. Capacity tests reserve a <b>10% provisional on-demand headroom</b> - 40 extra requests/hour initially and 200/hour at growth load - because the scenario requires on-demand use but supplies no workload. The headroom is a <b>DESIGN ASSUMPTION</b> to replace after workflow observation. |
| Feature lookback      | Six hours; enough to provide recent trends while bounding online computation, subject to validation against alternatives.                                                                                                                                                                                                                                                                                                                                  |
| Central data envelope | Approximately 256 bytes per motor-second for normalized scalar data and 4 KiB per                                                                                                                                                                                                                                                                                                                                                                          |

| Planning item | Provisional value and justification                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Retention     | <p>motor-minute for vibration feature packets; these are sizing envelopes, not sensor specifications.</p> <p>90 days hot and two years cold for curated telemetry, labels, predictions, audit data, and the immutable feature vector/source references needed to reconstruct a prediction. Select raw high-rate vibration windows are retained under investigation policy. These are <b>DESIGN ASSUMPTIONS</b> subject to site policy and measured capacity.</p> |

## 2.2 Functional Requirements

| ID    | Requirement and measurable criterion                                                                                                                                                                                                                                | Classification / rationale                                                                             |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| FR-01 | For every valid window, invoke exactly one versioned primary model and record the 24-hour horizon.                                                                                                                                                                  | <b>DESIGN DECISION.</b><br>Preserves the fixed target and one-model scope.                             |
| FR-02 | Accept a versioned contract containing motor/sensor identity, event and ingestion timestamps, units, quality flags, and recent vibration, temperature, current, and RPM. Contract tests reject unknown schemas, identities, units, timestamps, or malformed values. | <b>DESIGN DECISION.</b><br>Ensures correct asset/time association and speed-aware processing [2], [3]. |
| FR-03 | A valid result contains class, probability in $[\emptyset, 1]$ , motor ID, prediction time, horizon, source-window interval, model/feature versions, and quality state. Invalid or stale input returns Insufficient Data, never Normal.                             | <b>DESIGN DECISION.</b> Makes results actionable and auditable.                                        |
| FR-04 | Score every 15 minutes and on authenticated demand.                                                                                                                                                                                                                 | <b>DESIGN ASSUMPTION.</b><br>Establishes a measurable                                                  |

| ID    | Requirement and measurable criterion                                                                                                                                                                                                                                                   | Classification / rationale                                                                                                                                                                             |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | Requests with the same motor, window, and model version are idempotent and rate-limited so they cannot displace due scheduled work. Tests expect 400 scheduled plus up to 40 on-demand predictions/hour initially and 2,000 plus up to 200/hour at growth load, all subject to NFR-01. | cadence, burst allowance, and load without claiming observed demand.                                                                                                                                   |
| FR-05 | Provide versioned interfaces among gateway, ingestion, storage, feature/training pipeline, inference, dashboard/alerts, and a CMMS/EAM-equivalent outcome source. An integration test traces telemetry to storage, prediction, display, and linked outcome.                            | <b>DESIGN DECISION.</b><br>Connects acquisition, processing, and maintenance action [1].                                                                                                               |
| FR-06 | Let the Maintenance Engineer rank/filter by motor, site/area, risk state, data status, and time; view probability, recent trends, freshness and versions; and open the maintenance record. No user or prediction may autonomously shut down a motor.                                   | <b>DESIGN DECISION.</b> Retains a human decision boundary and assigns every required interaction to the dashboard.                                                                                     |
| FR-07 | Use a frozen episode state machine. A valid score at or above entry threshold $T_{\text{enter}}$ starts one episode. Further high-risk scores update it. It closes for metric purposes only after two consecutive valid scores at or below exit threshold                              | <b>DESIGN ASSUMPTION/DECISION.</b><br>The two-score rule equals 30 minutes at the provisional cadence and limits chatter; thresholds and the complete state machine are frozen before test evaluation. |

| ID    | Requirement and measurable criterion                                                                                                                                                                                                                                | Classification / rationale                                                                                   |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
|       | T_exit, where T_exit < T_enter; Insufficient Data neither closes nor re-arms it. A later entry crossing starts a new episode. Dashboard workflow state is separate and records acknowledgement, assignment, comment, investigation, escalation, and closure reason. |                                                                                                              |
| FR-08 | Link immutable predictions to alert disposition, maintenance action, confirmed failure time/type, and false-alarm feedback. Only authorized review can approve an outcome for training.                                                                             | <b>DESIGN DECISION.</b><br>Supports delayed performance evaluation without retrospective prediction changes. |

## 2.3 Non-Functional Requirements

| ID / concern       | Proposed requirement and acceptance criterion                                                                                                                                                           | Classification / failure behavior                                                                                                                  |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-01 Performance | At initial and growth load, p95 time from scheduled window close or accepted on-demand request to dashboard visibility shall be ≤2 minutes; normal-path maximum shall be <15 minutes.                   | <b>PROPOSED DESIGN TARGET.</b> Minute-scale response is appropriate to an advisory 24-hour horizon; accepted on-demand work shares the same bound. |
| NFR-02 Throughput  | Sustain 100/500 scalar records per second, 100/500 vibration packets per minute, and 440/2,200 total predictions per hour, including the provisional on-demand allowance. A one-hour test shall show no | <b>DESIGN ASSUMPTION.</b><br>Derived from the stated envelope plus 10% headroom; actual demand must replace it after workflow observation.         |

| ID / concern                | Proposed requirement and acceptance criterion                                                                                                                                                                                                                                                                                                                                                                                        | Classification / failure behavior                                                                                                          |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
|                             | unexplained loss, unbounded queue growth, or latency breach.                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                            |
| NFR-03 Scalability          | Stateless ingestion, feature, and inference workers scale horizontally by motor-ID partition; stores expand through capacity and retention controls. Scaling uses queue lag, feature freshness, latency, and saturation.                                                                                                                                                                                                             | <b>DESIGN DECISION.</b> Preserves per-motor ordering while enabling parallel work.                                                         |
| NFR-04 Reliability          | Target $\geq 99.5\%$ monthly prediction-path availability: eligible scheduled opportunities whose result is persisted and retrievable before the next 15-minute cycle divided by all eligible scheduled opportunities, excluding approved maintenance before both numerator and denominator are calculated. Use health checks, bounded retry, idempotent writes, durable queues, backups, and the last approved bundle for rollback. | <b>PROPOSED DESIGN TARGET.</b> Opportunity-level measurement matches the scoring cadence and avoids an ambiguous minute-based denominator. |
| NFR-05 Graceful degradation | Missing, corrupt, or stale required data shall use only a model validated for that pattern or return Insufficient Data. A low-rate channel is provisionally stale after two scoring intervals (30 minutes).                                                                                                                                                                                                                          | <b>DESIGN ASSUMPTION.</b> Missing telemetry cannot become healthy evidence.                                                                |
| NFR-06 Connectivity         | Buffer the central-transmission envelope locally for 24 hours, expose utilization, replay original                                                                                                                                                                                                                                                                                                                                   | <b>DESIGN ASSUMPTION.</b> The duration matches the prediction horizon; it is not an observed outage                                        |



| ID / concern           | Proposed requirement and acceptance criterion                                                                                                                                                                                                                                                            | Classification / failure behavior                                                                                                                                 |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | event time, deduplicate, and prevent expired replay from creating a current alert.                                                                                                                                                                                                                       | requirement.                                                                                                                                                      |
| NFR-07 Maintainability | Version code, configuration, schema, features, models, and alert rules; retain each prediction's immutable feature vector and source references for two years so it can be reconstructed within that declared audit window. Unit, contract, data, integration, smoke, and rollback checks gate releases. | <b>DESIGN DECISION/ASSUMPTION.</b> Production ML requires controls beyond offline accuracy [9]; the audit duration follows the provisional cold-retention policy. |
| NFR-08 Monitoring      | Monitor sensor/data health, service/pipeline health, feature/input distributions, output/alert behavior, and delayed outcome performance; every alert has an owner, history, route, and runbook.                                                                                                         | <b>DESIGN DECISION.</b> Covers the production layers identified in ML readiness work [9].                                                                         |
| NFR-09 Retraining      | Review readiness monthly and when validated quality, drift, label-volume, or performance signals fire. A signal starts investigation, not automatic promotion; every candidate repeats validation and approval.                                                                                          | <b>DESIGN ASSUMPTION/DECISION.</b> Input change alone does not prove concept drift [10].                                                                          |
| NFR-10 Security        | Apply authenticated service identities, least privilege, encryption in transit/at rest, secret rotation, audit logging, OT/IT segmentation, and no inference-to-actuator path.                                                                                                                           | <b>DESIGN DECISION.</b> OT controls must account for performance, reliability, safety, topology, and security [8].                                                |

## 2.4 Data Requirements

### 2.4.1 Sources, provenance, and labels

The primary production source is time-series vibration, temperature, current, and RPM from motors through an industrial gateway. Each observation carries motor and sensor IDs, event and ingestion timestamps, units, schema and source versions, payload/value reference, and quality status. Additional load, torque, ambient, motor-type, or duty variables require documented provenance and explicit approval.

The second essential source is an authoritative maintenance/work-order or failure-event system. This source is a **DESIGN ASSUMPTION** until the plant identifies it. Maintenance and operations must approve what constitutes failure, its timestamp, confirmation status, and planned/unplanned status before labels are generated. For prediction time  $t$ , a positive label means an approved failure occurs in  $(t, t + 24 \text{ hours}]$ ; a negative requires full observation through the horizon and no approved failure. Planned shutdowns, ambiguous/duplicate events, already-failed states, post-maintenance stabilization, and intervention-affected outcomes are excluded or explicitly censored. Features may use only information available at or before  $t$ .

Paderborn and CWRU data [5], [6] may support preprocessing and code prototypes with provenance retained, but shall not be used as evidence of native 24-hour labels, plant prevalence, or achieved performance.

### 2.4.2 Quality and missing/corrupt readings

Validation checks schema, type, identity, unit, configured/physical range, timestamp plausibility, clock skew, duplicate key, ordering, continuity, and cross-channel alignment. Invalid records are quarantined with reason codes; corrections never overwrite the raw observation. Interpolation or imputation records the method and quality indicator, is fitted on training data only, and is applied by the same versioned package in training and serving.

A provisional six-hour inference window requires at least 95% of expected observations in each required low-rate channel, at least one valid vibration feature packet in the latest 15 minutes, and compliance with the 30-minute staleness rule. These are **DESIGN ASSUMPTIONS**, not universal sensor-quality standards. Boundary and fault-injection tests verify that incomplete, corrupt, out-of-range, duplicate, out-of-order, and stale inputs are quarantined or return Insufficient Data with a visible reason.

Dataset manifests retain source partitions, validation output, transformation and label-policy versions, and extraction time. Training, validation, and test partitions are separated by time and motor or motor group where feasible. Overlapping windows from one source recording cannot cross partitions.

### 2.4.3 Privacy, confidentiality, and security

As a **DESIGN DECISION pending site classification**, the system treats motor telemetry, predictions, model artifacts, and maintenance records as operationally sensitive. User identity is retained only for acknowledgement, assignment, approval, and audit. Role- and purpose-based access, encryption, export controls, retention enforcement, deletion, secret management, and access logging apply to hot, cold, backup, and derived data. Unnecessary personal notes are not collected. This design does not assert a plant-specific legal regime; a site review must establish any additional privacy, safety, regulatory, or retention obligations.

### 2.4.4 Expected data volume

Using the explicit planning envelope, 100 motors produce approximately:

- scalar data:  $100 \times 86,400 \times 256 \text{ bytes} = 2.21 \text{ GB/day}$ ;
- vibration feature data:  $100 \times 1,440 \times 4 \text{ KiB} = 0.59 \text{ GB/day}$ ; and
- combined logical volume: approximately  $2.80 \text{ GB/day}$ ,  $252 \text{ GB/90 hot days}$ , and  $2.05 \text{ TB/two cold years}$ .

At 500 motors, the corresponding logical envelope is approximately  $14.0 \text{ GB/day}$ ,  $1.26 \text{ TB hot}$ , and  $10.2 \text{ TB cold}$ . These **DESIGN ASSUMPTIONS** exclude replication, indexes, compression, object metadata, selectively uploaded raw vibration, retained feature vectors/source manifests, and filesystem overhead. Capacity review must measure those additions explicitly and compare actual growth with the envelope before storage exhaustion.

## 3. Goals and Metrics

### 3.1 Goal Hierarchy and Measurement Policy

The hierarchy is Business Goal → System Goal → Model Goal → Metric → Baseline → Target. Business outcomes are measured against the same motors' most recent 12 usable pre-pilot months, normalized by motor operating hours. The 12-month baseline and a minimum six-month pilot are **DESIGN ASSUMPTIONS**; if insufficient events exist, observation continues rather than manufacturing a result. A staggered rollout or matched non-pilot comparison should be used where practical to reduce confounding by workload, motor mix, season, staffing, and maintenance-policy changes.

No deployed-system baseline exists. Load, integration, and fault-injection tests establish the first system measurements. Model candidates are compared on identical time- and asset-separated data against Always Normal, Always High Risk, a maintenance-approved deterministic rule, and logistic regression. Public benchmark scores are not substituted for project baselines.

| Hierarchy                                                                                                          | Metric and baseline                                                                                                                                                                                                                                                                                                                                                           | Target / acceptable threshold                                                                     | Status                                                  |
|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| BG-01 reduce unplanned downtime → SG timely/available/valid predictions → MG detect failures                       | BM-01: unplanned motor downtime hours per 1,000 operating hours; historical same-motor baseline.                                                                                                                                                                                                                                                                              | ≥15% relative reduction.                                                                          | <b>PROPOSED DESIGN TARGET</b> , not an achieved result. |
| BG-02 reduce emergency burden without shifting cost → SG alert/data reliability → MG recall with actionable alerts | BM-02 emergency labor-hours and BM-03 total motor-maintenance cost per 1,000 operating hours.                                                                                                                                                                                                                                                                                 | Emergency labor-hours ≥10% lower; total cost no more than 5% above baseline while BM-01 improves. | <b>PROPOSED DESIGN TARGET.</b>                          |
| BG-03 avoid unnecessary work and fatigue → SG deduplicated alert workflow → MG control false alerts                | BM-04 = ML-initiated completed interventions closed with no actionable condition divided by all ML-initiated completed interventions with a recorded disposition. Use a historical comparator only if existing condition-triggered work has the same definition; otherwise report “no comparable baseline” and use the absolute pilot guardrail without claiming improvement. | ≤20%; model alert precision/FPR gates below must also pass.                                       | <b>PROPOSED DESIGN TARGET.</b>                          |

Metric governance follows the decision hierarchy. The Plant Operations Manager owns BM-01 and the BM-03 cost guardrail; the Maintenance Engineer owns BM-02 and BM-04 disposition quality; the ML/Platform Engineer owns SM-01-SM-05 and produces MM-01-MM-06 release evidence for joint maintenance/operations approval. Business metrics use the minimum six-month pilot and a rolling 12-month view when enough outcomes exist.

System metrics are reviewed continuously and summarized monthly, except load/reconciliation acceptance tests, which run per release. Model metrics are calculated per candidate on the frozen untouched test set and then on matured operational outcomes; counts, ratios, uncertainty intervals, and relevant motor/operating-regime slices are reported together. Thus model quality is necessary for timely and actionable alerts, system quality is necessary for the model output to reach users, and neither alone establishes business benefit.

The numerical values are **initial negotiation targets**, not literature-derived norms. The business targets require a material pilot benefit while the 5% cost and 20% no-actionable-condition limits prevent apparent downtime improvement from being purchased through uncontrolled maintenance. The model targets make recall primary but impose precision and motor-day false-positive constraints so the implied alert workload is not unlimited;  $F1 \geq 0.63$  is the rounded minimum implied by recall 0.85 and precision 0.50.  $PR-AUC \geq 2 \times$  held-out prevalence requires a material improvement over the no-skill ranking baseline, while Brier improvement requires probabilities to outperform a constant-prevalence predictor. Before model development and before viewing the untouched test set, maintenance and operations must approve or revise these values using measured failure prevalence, error cost, intervention capacity, and shadow-mode alert volume. If those inputs are unavailable, the values remain provisional gates and cannot be represented as organizationally validated tolerances.

### 3.2 System Goals and Metrics

| ID    | Metric and measurement point                                                                                                                                                                                                                                                      | Baseline                    | Proposed target                                                                                                                      |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| SM-01 | For each valid new/updated High Risk episode, measure both durable dashboard persistence and successful handoff to the configured notification connector; a retry updates the same episode and does not count as a new alert. Human reading is not claimed as technical delivery. | End-to-end acceptance test. | $\geq 99\%$ persisted and handed off within five minutes of window close; zero duplicate active episodes for the same motor/horizon. |
| SM-02 | Scheduled window                                                                                                                                                                                                                                                                  | First representative        | $p95 \leq 2$ minutes and                                                                                                             |

| ID    | Metric and measurement point                                                                                                                                                                                                                                                                                                  | Baseline                                    | Proposed target                                                                                                                                                       |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | close or accepted on-demand request to dashboard visibility, reported at p50/p95/p99/maximum.                                                                                                                                                                                                                                 | load test including 10% on-demand headroom. | normal-path maximum <15 minutes at the 440/2,200 total hourly prediction envelopes.                                                                                   |
| SM-03 | Eligible scheduled opportunities whose result is persisted and retrievable before the next scoring cycle divided by all eligible scheduled opportunities; approved maintenance is removed from numerator and denominator.                                                                                                     | First measured pre-production/pilot month.  | ≥99.5% per calendar month.                                                                                                                                            |
| SM-04 | Reconcile every application-level gateway message ID to exactly one durable raw record and one terminal disposition - valid, quarantined with reason, or explicitly pending - after deduplication. Separately measure valid-data processing delay. MQTT broker acknowledgement is transport state, not durable-storage proof. | Test manifest and first pilot month.        | 100% terminal reconciliation in one-hour load tests; ≥99.5% of valid records available for processing within five minutes in normal operation; zero unexplained loss. |
| SM-05 | Valid predictions divided by eligible                                                                                                                                                                                                                                                                                         | First shadow-mode month.                    | ≥95% valid coverage; 100% of                                                                                                                                          |

| ID | Metric and measurement point                                  | Baseline | Proposed target                                                |
|----|---------------------------------------------------------------|----------|----------------------------------------------------------------|
|    | scheduled motor-windows; invalid attempts carry reason codes. |          | non-valid attempts visibly classified, not returned as Normal. |

### 3.3 Model Goals, Evaluation Units, and Metrics

A scoring window is one valid prediction opportunity. Alert episodes follow the frozen FR-07 state machine: entry at T\_enter, updates while active, closure after two consecutive valid scores at or below T\_exit, no state change on Insufficient Data, and re-entry only after closure. For evaluation, approved failures are processed chronologically and matched one-to-one to the earliest unmatched episode starting in the preceding 24 hours; other episodes are unmatched. A continuously active episode that began more than 24 hours before a later failure is not retrospectively moved into the target window.

A negative motor-day is an adequately observed motor-day with no approved failure in the following 24 hours and no censoring condition.  $FP\_day = 1$  when at least one new unmatched episode starts on that day; an episode continuing from a prior day is not counted again.  $TN\_day = 1$  when no new unmatched episode starts. Post-deployment episodes followed by preventive intervention are censored unless an approved outcome policy supports attribution; they are not automatically false positives. Every baseline and candidate is simulated through this identical state machine and matching policy.

| ID    | Metric                                                                                                                     | Baseline                                                                                                                         | Proposed promotion target                                              |
|-------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| MM-01 | Failure-event recall<br>$TP / (TP + FN)$ .                                                                                 | Always Normal = 0;<br>rule and logistic<br>baselines measured<br>on the same test set.                                           | $\geq 85\%$ .                                                          |
| MM-02 | Alert-episode<br>precision<br>$TP / (TP + FP)$ .                                                                           | Always High Risk<br>equals prevalence<br>induced by the<br>episode policy.                                                       | $\geq 50\%$ .                                                          |
| MM-03 | $F1\ 2PR / (P + R)$ .                                                                                                      | Always Normal = 0;<br>practical baselines<br>measured.                                                                           | $\geq 0.63$ while separate<br>recall and precision<br>gates also pass. |
| MM-04 | Negative-motor-day<br>false-positive rate<br>$FP\_day / (FP\_day + TN\_day)$ under the<br>stated new-episode<br>indicator. | Always Normal = 0;<br>simulate and report<br>Always High Risk,<br>rule, and logistic<br>policies under the<br>same state machine | $\leq 1\%$ .                                                           |

| ID    | Metric                                                                                                        | Baseline                                                            | Proposed promotion target                                                                                                                                          |
|-------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |                                                                                                               | rather than asserting a universal value.                            |                                                                                                                                                                    |
| MM-05 | Window-level area under the precision-recall curve (PR-AUC).                                                  | Held-out positive-window prevalence plus rule/logistic comparators. | $\geq 2\times$ held-out prevalence and above both practical baselines; report a 95% bootstrap interval.                                                            |
| MM-06 | Overall Brier score, with mandatory reliability plots and counts by predeclared motor/operating-regime slice. | Constant training-prevalence probability on the same test set.      | Overall Brier score lower than the constant baseline. Slice plots are mandatory diagnostics with written disposition, not an unsupported numerical promotion gate. |

PR-AUC is included because the operational question is whether positive predictions remain useful as recall increases in an imbalanced problem; accuracy is excluded because an Always Normal policy can appear strong while detecting no failures. Recall receives priority because a false negative leaves an impending failure without an ML warning. Apply the frozen episode policy to validation data; reject every threshold with recall <85%, precision <50%, FPR >1%, or F1 <0.63; and select the highest-recall survivor, using expected alert volume, lead time, and Brier/calibration evidence as secondary comparisons. Freeze the selected threshold, exit threshold, state machine, and matching policy before untouched testing. If no candidate passes every mandatory gate, do not promote it.

### 3.4 Acceptance and Promotion

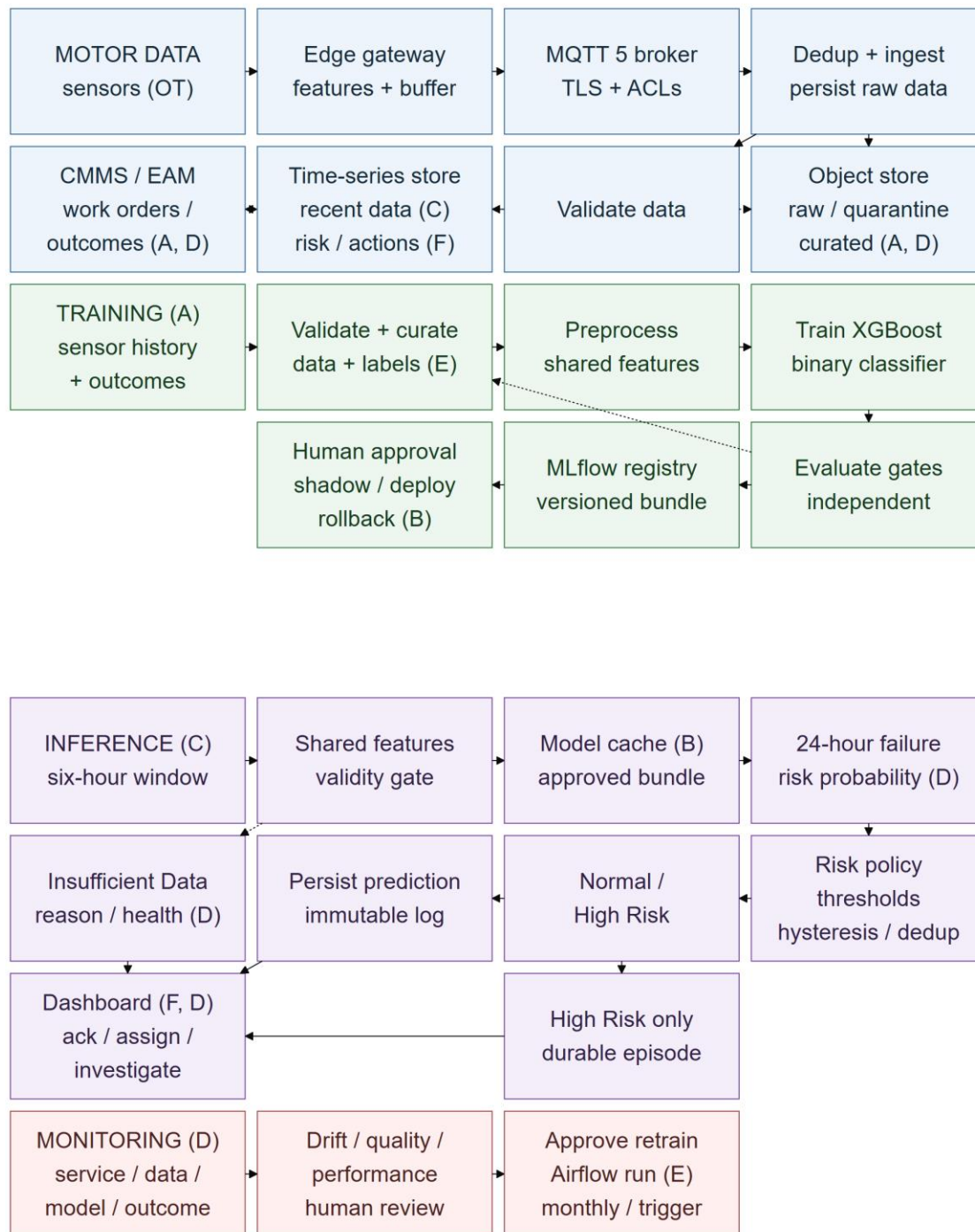
Promotion requires an approved failure definition; leakage-resistant time/asset partitions; all numerical MM-01-MM-06 gates on untouched data; the mandatory MM-06 slice diagnostics and dispositions; representative SM-01-SM-05 integration, load, and fault tests; denominators, uncertainty, and important slices; versioned model/data/feature/threshold/episode artifacts; monitoring and rollback readiness; and human approval. Monitoring evidence is required for promotion with no grace exception; an authorized grace period may apply only to continued serving of the current approved bundle. Business metrics are pilot exit criteria, not prerequisites for shadow mode. A target revision must be approved before the untouched test set is examined. No threshold in this section represents achieved performance.



## 4. High-Level Architecture Design

### 4.1 Architectural Stance and Diagram

The proposed deployment is a small central on-premises VM/container cluster connected to the OT area through industrial gateways and an authenticated MQTT broker. Gateways perform acquisition, identity/time/unit normalization, versioned high-rate vibration feature extraction, and durable buffering. Central services validate, store, train, register, score, monitor, and deliver alerts. Training and serving share the same versioned feature package. One XGBoost binary classifier is the primary predictive model; deterministic rules and logistic regression remain non-production comparators.



*High-level training, inference, and feedback architecture*

**Figure 1. Proposed production ML architecture.** Four bands separate data acquisition/storage, training, inference, and monitoring/retraining. Matching letters show transfers: A, curated history and approved outcomes to training; B, approved bundle or rollback to inference; C, validated recent data to inference; D, data health, serving telemetry and matured outcomes to monitoring; E, approved retraining back to batch

validation; F, predictions, alert state and workflow actions back to the operational store. Solid arrows show routine flows; dotted arrows show invalid-data or failed-evaluation paths. The architecture is prospective and does not represent an existing deployment.

## 4.2 Data Flow and ML Pipeline

### 4.2.1 Training path

sensor/history data → ingestion and immutable storage → batch validation and curation → shared preprocessing/feature engineering → XGBoost training → independent evaluation → MLflow registry → human-approved deployment

Gateways publish scalar envelopes and vibration feature packets through MQTT over TLS using the selected at-least-once path. MQTT acknowledgement records transport state only. Ingestion authenticates and deduplicates each application message ID, durably lands one raw record, assigns a valid/quarantined/pending disposition, and then emits a separate application-level durable receipt to the gateway. Batch validation quarantines records that fail identity, schema, timestamp, unit, range, continuity, or lineage rules. The CMMS/EAM adapter joins approved outcomes using the governed ( $t$ ,  $t+24$  hours] policy. The shared feature package builds the six-hour window and fits any learned transformation on training data only. Airflow creates immutable dataset manifests and time/asset splits, trains the single candidate, and runs independent evaluation. A failed mandatory gate prevents registry promotion. Passing bundles contain the model, transformations, schema, threshold/episode policy, evaluation evidence, and lineage. Human approval, compatibility checks, shadow operation, and rollback readiness precede activation.

### 4.2.2 Inference path

new sensor data → MQTT ingestion → online validation and recent storage → shared preprocessing/features → cached approved model → failure-risk probability → versioned threshold/hysteresis → Normal/High Risk → dashboard/alert

Every 15 minutes, or on authorized demand within the reserved capacity, C15 retrieves a six-hour motor window and invokes the shared C09 preprocessing/feature package. Missing, stale, corrupt, or incompatible data persist Insufficient Data with a reason and data-health event; they do not create or update a risk episode. Otherwise the cached approved XGBoost bundle returns one probability. C16 applies the frozen threshold and episode state machine to produce Normal or High Risk. The system persists the immutable prediction; only a high-risk transition/update creates or updates a durable risk episode. C17 then presents the result and hands an eligible episode to the configured notification connector. The engineer can inspect, acknowledge, assign, investigate, close with a reason, and link a work order. No inference result actuates equipment.

### 4.2.3 Feedback and MLOps path

service/data/model/outcome monitoring → validated quality, drift, or performance signal → human investigation → approved retraining trigger →

validation/training/evaluation → registry → shadow/approved deployment or rejection

Monitoring covers gateway health, missing/stale data, validation failures, queue lag, feature freshness, latency, availability, prediction distribution, episode rate, drift indicators, and delayed performance/calibration. Distribution change opens an investigation; it is not automatic proof of concept drift [10]. Monthly review and validated signals can start the controlled Airflow path, but no training process can directly update serving. Alert dispositions and CMMS outcomes are retained immutably, and missing outcomes never become negative labels.

### 4.3 Component Responsibilities, Technologies, and Failure Behavior

| ID / component             | Responsibility; input → output                                                                                                                             | Candidate technology and rationale                                                   | Important failure behavior                                                                                                                                                   |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C01 Sensors/asset registry | Observe vibration, temperature, current, RPM and bind channel metadata → timestamped readings.                                                             | Existing plant-grade sensors and asset registry; avoids a vendor assumption.         | Invalid/unregistered channels become unavailable, not healthy defaults.                                                                                                      |
| C02 Edge gateway/buffer    | Normalize identity/time/unit, compute vibration features, buffer → scalar envelopes, feature packets, quality metadata.                                    | Industrial Linux, containerized Python/NumPy/SciPy, durable SQLite/filesystem queue. | Buffer 24-hour envelope, alarm on utilization, preserve original timestamps.                                                                                                 |
| C03 MQTT broker            | Authenticate and decouple gateways/consumers → per-motor topics and delivery metrics.                                                                      | MQTT 5/EMQX; lightweight publish/subscribe with explicit delivery semantics [7].     | Gateways buffer outage; consumers deduplicate at-least-once delivery.                                                                                                        |
| C04 Ingestion/persistence  | Validate transport, deduplicate an application message ID, land one immutable raw record, assign its disposition, and emit a durable application receipt → | Python, Pydantic, MQTT/MinIO/PostgreSQL clients.                                     | MQTT PUBACK is not storage proof; the gateway retains work until the application receipt. Poison messages receive a durable quarantine disposition rather than disappearing. |

| ID / component                         | Responsibility; input<br>→ output                                                                                                                               | Candidate<br>technology and<br>rationale                                                          | Important failure<br>behavior                                                                                                                                                                                                                                  |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | validation work<br>item/audit.                                                                                                                                  |                                                                                                   |                                                                                                                                                                                                                                                                |
| C05<br>Validation/quarantine           | Apply stream/batch<br>quality rules → valid<br>normalized or<br>quarantined<br>records.                                                                         | Pydantic/Pandera<br>and batch validation<br>reports.                                              | Invalid data isolated<br>with reason; rules<br>failure stops the<br>partition.                                                                                                                                                                                 |
| C06 Historical store                   | Retain raw,<br>quarantine, curated,<br>manifests,<br>feature/source<br>snapshots, and<br>artifacts → versioned<br>objects under<br>measured lifecycle<br>rules. | On-premises<br>MinIO/S3 lifecycle<br>storage.                                                     | Failed/checksum-<br>invalid writes block<br>or quarantine;<br>training pauses if<br>unavailable. Raw<br>high-rate retention is<br>selective, while each<br>served prediction<br>retains the<br>replayable<br>feature/source<br>evidence required by<br>NFR-07. |
| C07 Operational<br>store               | Serve recent<br>windows,<br>predictions,<br>episodes, audit →<br>queryable state.                                                                               | PostgreSQL with<br>TimescaleDB; time-<br>window and<br>relational integrity at<br>moderate scale. | Transactional retry;<br>stale read-only view<br>is labelled; no fresh<br>score from stale<br>data.                                                                                                                                                             |
| C08 CMMS/EAM<br>adapter/labels         | Join work<br>orders/outcomes<br>and apply label<br>policy → versioned<br>labels/censoring.                                                                      | Authenticated<br>REST/file adapter<br>and Airflow<br>Python/SQL job.                              | Missing/ambiguous<br>outcomes remain<br>delayed/censored,<br>never negative by<br>default.                                                                                                                                                                     |
| C09 Shared feature<br>package          | Align modalities,<br>check validity,<br>transform →<br>versioned vector and<br>quality lineage.                                                                 | Versioned Python<br>package using<br>NumPy/SciPy/panda<br>s/scikit-learn<br>transformers.         | Version mismatch<br>fails training or<br>returns<br>Insufficient Data;<br>no online refit.                                                                                                                                                                     |
| C10<br>Orchestrator/dataset<br>builder | Schedule curation,<br>snapshots, splits,<br>jobs → immutable<br>manifest/run state.                                                                             | Apache Airflow;<br>suited to auditable<br>scheduled<br>dependencies.                              | Failed dependency<br>stops downstream<br>work; serving is<br>independent.                                                                                                                                                                                      |

| ID / component               | Responsibility; input<br>→ output                                                                                                                   | Candidate<br>technology and<br>rationale                                                                                                                | Important failure<br>behavior                                                                                                                         |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| C11 Primary training         | Fit one candidate → XGBoost artifact and metadata.                                                                                                  | Containerized Python/XGBoost, selected as a maintainable engineered-feature candidate whose latency and model quality still require acceptance testing. | Invalid features/classes or compute failure produces no candidate or deployment authority.                                                            |
| C12 Evaluation gate          | Calculate metrics, uncertainty, slices, baselines, contracts → signed pass/fail report.                                                             | Separate scikit-learn evaluation container and policy checks.                                                                                           | Missing lineage, leakage, failed metric, or schema incompatibility rejects promotion.                                                                 |
| C13 Registry                 | Retain immutable bundle, evidence, aliases, approvals → approved/candidate versions.                                                                | Self-hosted MLflow backed by PostgreSQL/MinIO.                                                                                                          | Outage blocks change, not cached serving; unsigned/corrupt artifact is rejected.                                                                      |
| C14 Deployment controller    | Human approval, smoke/shadow checks, atomic activation → running version/audit.                                                                     | CI runner, OCI registry, Docker on on-premises VMs, Ansible-equivalent rollout.                                                                         | Failed checks leave active version unchanged; regression rolls back.                                                                                  |
| C15 Feature/inference worker | Schedule scores, validate window, load cache → probability or Insufficient Data.                                                                    | Python worker, FastAPI for on-demand requests, XGBoost runtime, durable queue.                                                                          | Retry idempotently; only verified cached approved model may serve.                                                                                    |
| C16 Risk/episode policy      | Apply T_enter, T_exit, two-score closure, no-change-on-insufficient-data, deduplication, and event matching → class and durable episode transition. | Versioned Python state machine and PostgreSQL transaction/outbox.                                                                                       | Invalid probability cannot become Normal; configuration mismatch blocks classification; the frozen state-machine version is stored with every result. |

| ID / component                     | Responsibility; input<br>→ output                                                                                                                                                                                     | Candidate<br>technology and<br>rationale                                               | Important failure<br>behavior                                                                                                                                             |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C17<br>Dashboard/notification      | Present/filter by motor, site/area, risk, data status, and time; show trends/freshness; record acknowledgement, assignment, comment, investigation, escalation, and closure reason → audited action and CMMS linkage. | Internal web application, API, PostgreSQL workflow, configurable notifier.             | Persist before notification handoff; record connector acknowledgement separately; queue outage recovery; deny unauthorized state/control action.                          |
| C18<br>Observability/ML monitoring | Aggregate service, data, input, output, outcome signals → dashboards, alerts, reports.                                                                                                                                | Prometheus/Grafana, structured logs, scheduled Python/SQL monitors.                    | Independent heartbeat; monitoring loss blocks promotion immediately. Only continued serving of the current approved bundle may use an explicitly authorized grace period. |
| C19 Retraining controller          | Convert validated signal and owner decision → approved/rejected Airflow trigger.                                                                                                                                      | Ticket/approval workflow and Airflow trigger API.                                      | Drift alone cannot deploy; missing labels/approval leaves serving unchanged.                                                                                              |
| C20 Security/audit controls        | Enforce identity, segmentation, encryption, secrets, audit → permitted flows and evidence.                                                                                                                            | Existing IdP/OIDC, PKI/mTLS, Vault-compatible secrets, firewall/DMZ, encrypted stores. | Fail closed; audit failure blocks privileged promotion, never motor operation.                                                                                            |

These products are candidates, not claims about existing plant technology. Equivalent standard tools may replace them if input/output contracts, controls, traceability, and failure semantics remain unchanged. The selected MQTT at-least-once path requires application-level deduplication/idempotency; broker acknowledgement alone is not proof

of durable raw landing, prediction persistence, or notification handoff [7]. OT placement and security must respect reliability, safety, and performance constraints [8].

## 4.4 Deployment, Scaling, Reliability, and Maintenance

The gateway tier uses one industrial gateway per suitable motor group. Stateless central workers scale horizontally by motor-ID partition and durable queues; TimescaleDB and MinIO use backup/replication and capacity controls. Training runs in bounded containers with inference capacity reserved. The **DESIGN DECISION** is to begin without a full Kubernetes estate or public-cloud dependency because the assumed 100/500-motor envelope can be tested on the smaller VM/container design; representative load, resilience, and rollback tests must confirm that judgment.

During gateway/network loss, data buffer locally and later replay with original event time. Broker/ingestion loss causes bounded retry. Object-store loss pauses new durable landing and training; registry loss blocks release but cached approved serving continues. Operational-store loss prevents claiming new persisted predictions; the dashboard may show a clearly stale last-known view. Dashboard loss preserves durable alerts for replay. CMMS loss delays outcomes and performance measurement. Model/schema incompatibility fails closed. Candidate failure leaves the active model unchanged. These behaviors provide graceful degradation without converting infrastructure failure into Normal motor status.

Maintainability relies on immutable dataset and model lineage, shared transformation code, independent evaluation, human-gated registry aliases, shadow rollout, rollback, layered monitoring, and controlled retraining. This treats production readiness as a property of the complete system rather than of one offline score [9].

## 5. Trade-offs Analysis

### 5.1 TO-01 - False Negatives versus False Positives

**Engineering question.** How should the operating threshold balance missed 24-hour failures against false alerts and unnecessary maintenance?

- **Option A - recall-first lower threshold:** improves the chance of warning before a genuine failure and supports downtime reduction, but increases investigation work, false episodes, cost, and alert fatigue.
- **Option B - precision-first higher threshold:** produces a smaller and more actionable queue, but misses more failures and can make the system quiet rather than useful.

**Scenario consequence and decision.** Operations bears much of the false-negative downtime cost, while maintenance bears false-positive workload. The **DESIGN DECISION** is constrained recall-first selection: under the frozen C16 entry/exit/closure and matching



state machine, maximize validation recall among thresholds satisfying recall  $\geq 85\%$ , alert-episode precision  $\geq 50\%$ , negative-motor-day FPR  $\leq 1\%$ , and F1  $\geq 0.63$ . C17 retains human review. These are provisional negotiation targets to approve against prevalence, error cost, and intervention capacity before untouched evaluation. If no threshold passes, do not deploy. Residual risks are uncertain plant prevalence and costs, rare-event uncertainty, slice disparities, and intervention-censored outcomes; monitor them through uncertainty, slice, alert-disposition, and business-cost reporting.

## 5.2 TO-02 - Edge versus Centralized/Cloud Inference

**Engineering question.** Should complete risk inference run on every gateway or in one governed central service?

- **Option A - full edge inference:** continues local scoring during central-path loss and reduces telemetry transport, but multiplies model rollout, resource, monitoring, rollback, and version-drift risks.
- **Option B - central inference:** provides one model/feature/threshold governance point and consistent fleet monitoring, but depends on the central path and creates a wider central failure domain. Public cloud would additionally introduce an unapproved OT boundary.

**Scenario consequence and decision.** The 24-hour horizon and 15-minute cadence do not require millisecond control, while stakeholders need consistent fleet results and governed releases. The **DESIGN DECISION** is gateway-side collection, buffering, and vibration features with centralized on-premises C15 inference; neither a gateway risk model nor public-cloud inference is deployed initially. This supports maintainability, monitoring, promotion, and the 100/500-motor envelope. Residual risks are central outage/capacity and interruptions beyond the 24-hour buffer; expose lag/freshness, replay recoverable data, return Insufficient Data, and retain rollback. A future edge failover requires a separate validated design.

## 5.3 TO-03 - Prediction/Data Freshness versus Infrastructure Cost

**Engineering question.** How much freshness is justified for a 24-hour advisory decision?

- **Option A - central raw streaming and sub-minute scoring:** can show changes sooner and retain richer data, but increases bandwidth, storage, compute, monitoring, and alert churn without evidence of better performance for this target.
- **Option B - bounded aggregation and cadence:** lowers cost and supports stable windows and testable capacity, but can delay fast changes and discard raw-waveform information.

**Scenario consequence and decision.** Maintenance needs stable actionable episodes, operations needs bounded cost, and platform engineering needs a reproducible capacity envelope. The **DESIGN DECISION** is the assumed one-second scalar envelope, one-minute edge vibration packet, six-hour feature window, 15-minute/on-demand scoring, 90-

day hot and two-year cold retention, and selective raw-vibration upload. The p95 two-minute and normal-path <15-minute targets remain. Residual risks include fast-onset failures, an unsuitable lookback, and inaccurate payload/retention estimates. Pilot measurements of lead time, coverage, queue lag, storage growth, and missed-event timing must drive any versioned revision.

## 5.4 TO-04 - Model Complexity/Performance versus Latency and Maintainability

**Engineering question.** Should the primary model learn from raw temporal sequences or from governed engineered features?

- **Option A - temporal/deep model:** may learn patterns omitted by manual summaries, but requires more representative labelled histories, raw-data infrastructure, compute, tuning, monitoring, and diagnostic effort. No project evidence shows that it improves this exact target.
- **Option B - engineered features plus XGBoost:** supports fast bounded inference, shared feature lineage, calibration, and simpler rollback, but may miss temporal structure and depends on feature quality.

**Scenario consequence and decision.** The design has no verified plant dataset or achieved comparison; maintainability and evidence gates therefore matter more than speculative complexity. The **DESIGN DECISION** is one XGBoost primary classifier with deterministic and logistic baselines only. C09 owns the shared feature package, C11 trains the candidate, C12 evaluates it, C13 registers an approved bundle, C14 deploys it, and C15 serves it. It must pass MM-01-MM-06 and system gates before promotion. This follows the evidence that method choice must fit the data and operating problem [4] and avoids treating public benchmark transfer as proven [5]. Expected latency and maintenance benefits remain design hypotheses until load, rollback, and model-quality tests pass. Residual risk is that engineered features/XGBoost may fail performance or calibration gates. That outcome triggers data/label review and a new controlled comparison, possibly including a temporal candidate; it does not justify relaxing gates or silently adding a second production model.

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